HCI Week-3

CONCEPTUALIZING INTERACTION

design example 1

- A designer comes up with the idea of creating a robot waiter.
  - help waiters in a restaurant take orders and deliver meals to customers

Conceptualizing design example 1

Proof of concept

- Conceptualize what the proposed product will do

Why the need to conceptualizing design?

- To scrutinize vague ideas and assumptions about the benefits of the proposed product and user experience in terms of their feasibility
- How realistic is it to develop?
- How desirable and useful?

- Many unknowns need to be considered in the initial stages of a design project
- Especially if it is a new product that is being proposed

Robot Waiter

- Why ? What problem would this address?
- Putative benefits
  - Could help take orders and entertain customers by having a conversation with them at the table
- Design concerns
  - How would it need to move to appear to be talking?
  - What would the customers think of it?
  - Would it always be a pleasure for them to engage with the robot, not knowing what it would say on each new visit to the restaurant?
  - Would it be annoying or a fad?

Consider design example 2

- An automated robot server that is functional
- In contrast to the idea of a waiter robot the robot server addressed a real-world problem
  - the cooks load up the trays on its shelves, and then sends it to a preset area of the dining room to deliver the food
  - the customers remove the food and drinks from the robot server
- It is like a tool, there to save waiters' time rather than replacing them

Which type of server is preferable and on what grounds?

What is an assumption?

- Taking something for granted when it needs further investigation, e.g.
  - The robot could take orders and entertain customers by having a conversation with them
  - The robot could make recommendations for different customers, such as restless children or fussy eaters

What is a claim?

- A claim is stating something to be true when it is still open to question
  - It will raise revenue by attracting more people to the restaurant
  - It will solve the labor crisis of there not being enough human waiters
- The real problem being addressed:

  "It is difficult to recruit good wait staff who provide the level of customer service to which we have become accustomed."


DAWN
نیوز

Working through assumptions

- Need to work through the initial stages of a design project
  - Where do your ideas come from?
  - What sources of inspiration were used?
  - Is there any theory or research that can be used to inform them?
- During the early ideation process
  - Ask questions, reconsider assumptions, and articulate concerns

Poor assumptions

- Those that are difficult to articulate
  - can highlight what ideas are vague or unrealistic
  - identify human activities and interactivities that are problematic
- Based on this kind of conceptual analysis iteratively work out how the design ideas might be improved

A framework for analyzing the problem space

- Are there problems with an existing product or user experience? If so, what are they?
- Why do you think there are problems?
- How do you think your proposed design ideas might overcome these?
- If you are designing for a new user experience, how do you think your proposed design ideas support, change, or extend current ways of doing things?

Activity

- What were the assumptions and claims made about watching 3D TV?

Figure 3.2 A family watching 3D TV

Source: Andrey Popov, [Shutterstock]

Assumptions and claims: how realistic?

- There was no existing problem to overcome
  - What was being proposed was a new way of experiencing TV

- An assumption
  - People would really enjoy the enhanced clarity and color detail provided by 3D

- A claim
  - People would not mind paying a lot more for a new 3D-enabled TV screen because of the new experience

Benefits of conceptualizing

Orientation

- Enables design teams to ask specific questions about how the conceptual model will be understood

Open-minded

- Prevents design teams from becoming narrowly focused early on

Common ground

- Allows design teams to establish a set of commonly agreed terms

From problem space to design space

- Having a good understanding of the problem space can help inform the design space
  - for example, what kind of interface, behavior, functionality to provide
- Before deciding upon these, it is important to develop a conceptual model

Conceptual model

- A conceptual model is:

  “…a high-level description of how a system is organized and operates” (Johnson and Henderson, 2002, p26)

- It enables:

  “…designers to straighten out their thinking before they start laying out their widgets” (Johnson and Henderson, 2002, p28)

- Provides a working strategy and framework of general concepts and their interrelations

Components

- Metaphors and analogies
  - Understand what a product is for and how to use it for an activity
- Concepts that people are exposed to through the product
  - Task–Domain objects, their attributes, and operations (for example, saving, revisiting, organizing)
- Relationship and mappings between these concepts

First steps in formulating a conceptual model

- What will people be doing when carrying out their tasks?
- How will the proposed model support these?
- What kind of interface metaphor, if any, will be appropriate?
- What kinds of interaction modes and styles to use?
- Keep in mind when making design decisions how the user will understand the underlying conceptual model

Conceptual models

- Many kinds and ways of classifying them
- The best conceptual models are often those that appear:
  - Obvious and simple
  - The operations they support are intuitive to use

Interface metaphors

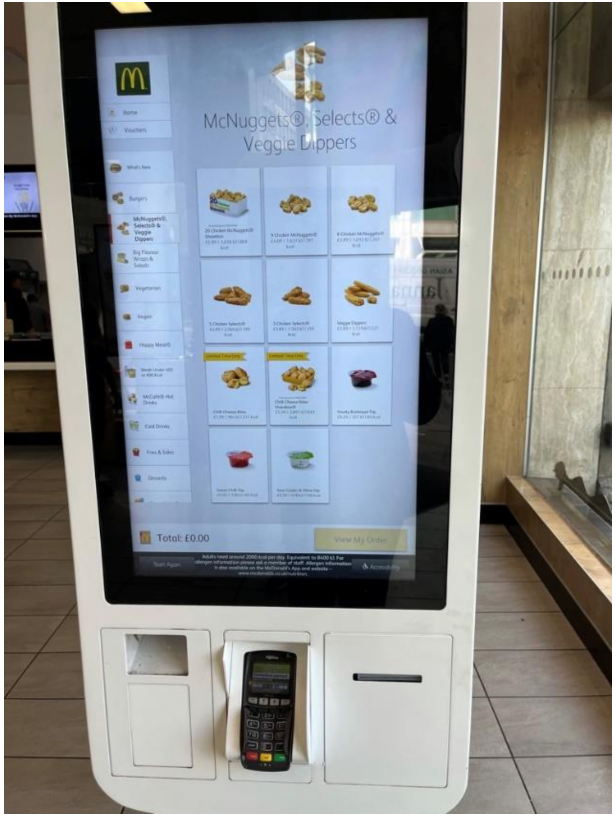
- Interface designed to be similar to a physical entity but also has own properties
  - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user’s familiar knowledge, helping them to understand ‘the unfamiliar’
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

Examples of interface metaphors

- Conceptualizing what users are doing
  - e.g, surfing the Web
- A conceptual model instantiated at the interface
  - e.g., the desktop metaphor
- Visualizing an operation
  - e.g., an icon of a shopping cart into which the user places items

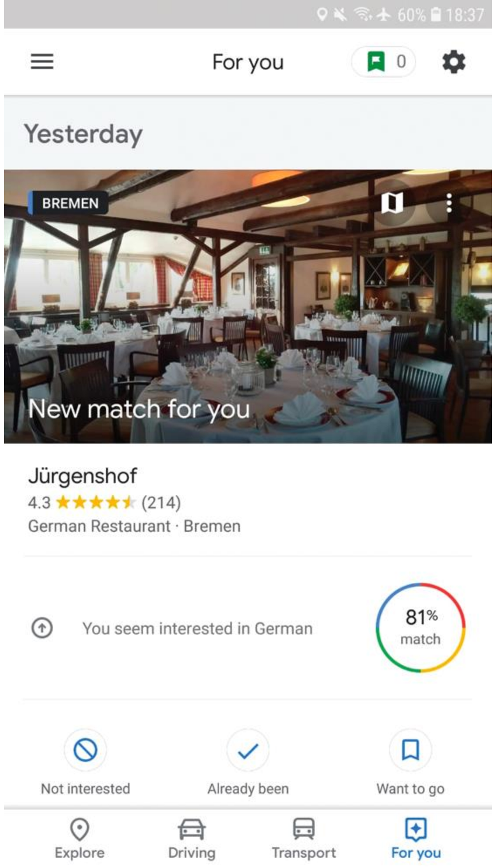
The card metaphor

- The card is a very popular metaphor. Why?
  - It has familiar form factor
  - It can easily be flicked through, sorted, and themed
  - Cards have strong associations, providing an intuitive way of organizing content that is “card sized
  - Structures content into meaningful chunks



A fast food display

Google Now card for restaurant



Benefits of interface metaphors

- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

Problems with interface metaphors

- Break conventional and cultural rules
  - e.g, recycle bin placed on desktop
- Can constrain designers in the way that they conceptualize a problem space
- Can conflict with design principles
- Designers can inadvertently use bad existing designs and transfer the bad parts over
- Limits designers' imagination in coming up with new conceptual models
- Often don't scale very well, especially when trying to describe how complex systems work (Cooper, 2020)

Activity

- Describe the components of the conceptual model underlying most online shopping websites, for example:
  - Shopping cart
  - Proceeding to check-out
  - 1-click
  - Gift wrapping
  - Cash register

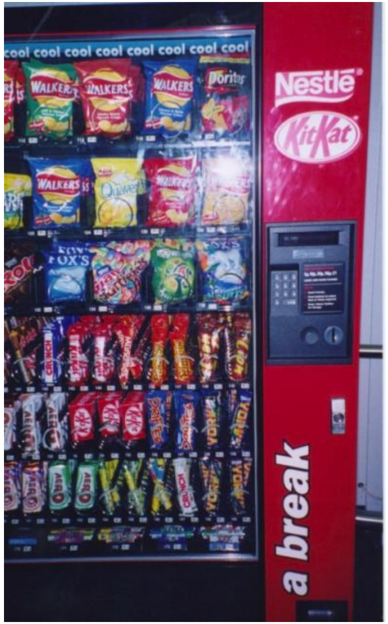
Interaction types

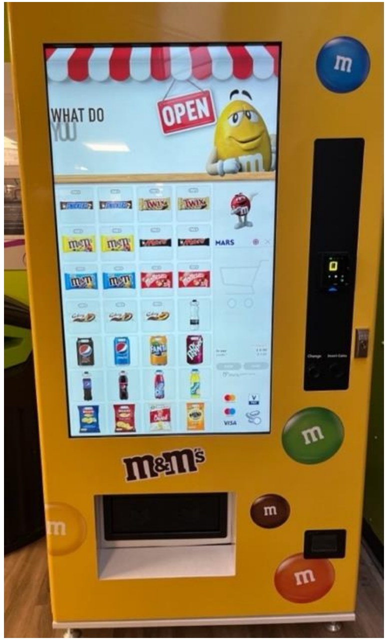
- Instructing
  - Issuing commands and selecting options
- Conversing
  - Interacting with a system as if having a conversation
- Manipulating
  - Interacting with objects in a virtual or physical space by manipulating them
- Exploring
  - Moving through a virtual environment or a physical space
- Responding
  - The system initiates the interaction and the person chooses whether to respond

1. Instructing

- Where someone instructs a system and tells it what to do
  - e.g.Tell the time, print a file, or save a file
- A very common conceptual model underlying a diversity of devices and systems
  - e.g. Word processors, vending machines
- The main benefit is that instructing supports quick and efficient interaction
  - Good for repetitive kinds of actions performed on multiple objects

Which is easiest and why?





2. Conversing

- Underlying model of having a conversation with another human
- Ranges from simple voice recognition menu-driven systems to more complex ‘natural language’ dialogs
- Examples include timetables, search engines, advice-giving systems, help systems, virtual agents, chatbots, toys, and pet robots designed to converse with people

Pros and cons of conversational model (1)

- Allows people to interact with a system in a way that is familiar to them
  - can make them feel comfortable, at ease, and less scared

- Misunderstandings can arise when the system does not know how to parse what someone says
  - e.g, voice assistants can misunderstand what children say

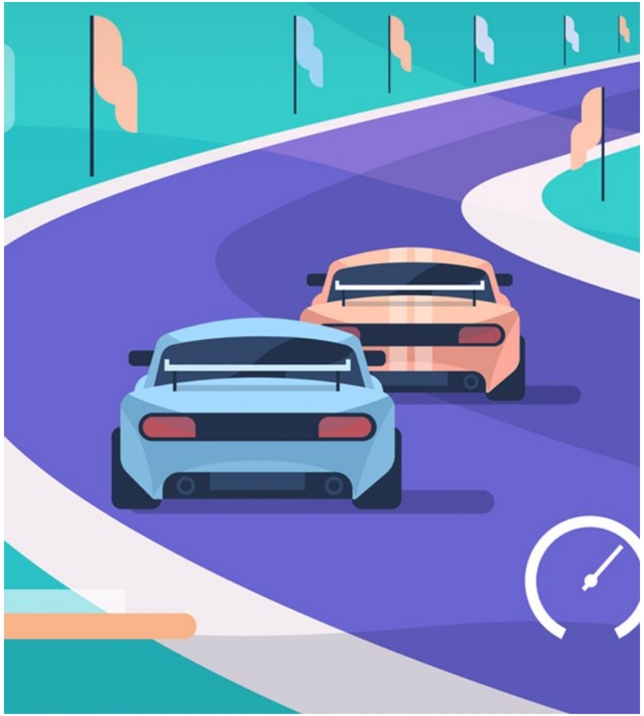
Siri's response when asked if a person needs an umbrella today

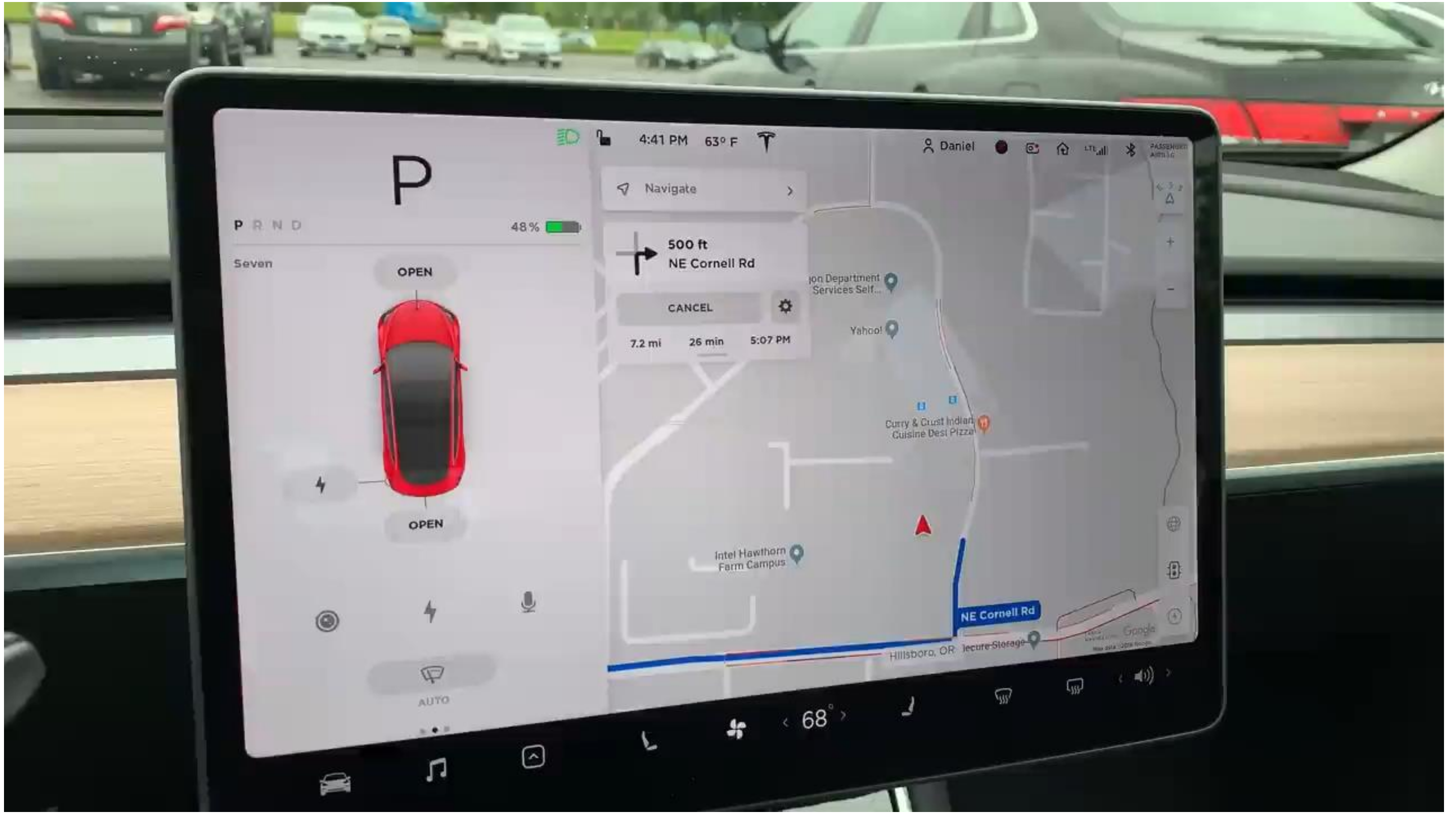
Pros and cons of conversational model (2)

"If you'd like to press 1, press 3.
If you'd like to press 3, press 8.
If you'd like to press 8, press 5..."

3. Manipulating

- Involves dragging, selecting, opening, closing and zooming actions on virtual objects
- Exploit's someone's knowledge of how they move and manipulate in the physical world
- Can involve actions using physical controllers or air gestures to control the movements of an on-screen avatar
- Tagged physical objects (for instance, balls) that are manipulated in a physical world result in physical/digital events (such as animation)

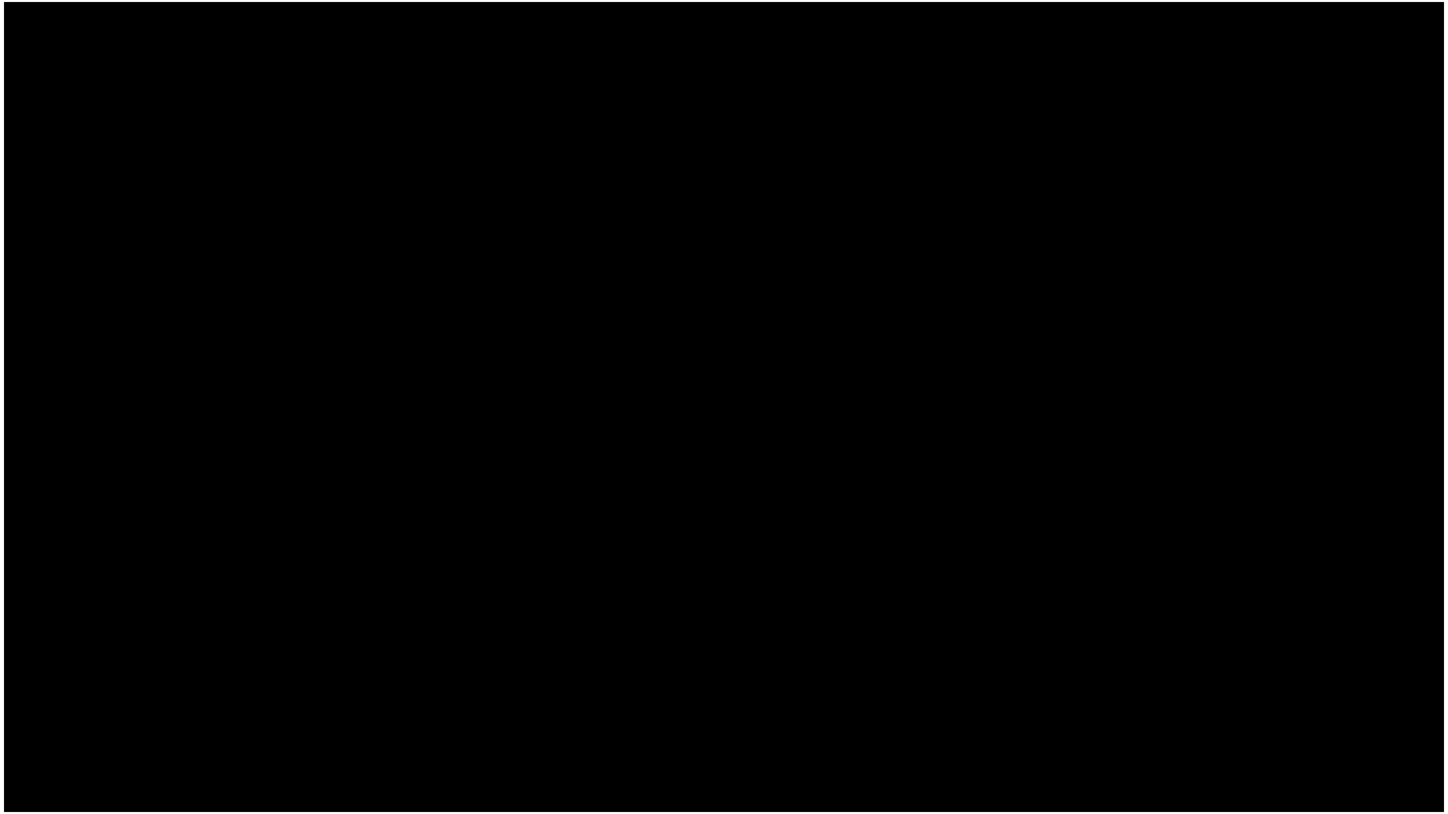




www.id-book.com

Direct Manipulation

- Ben Shneiderman (1983) coined the term
- Three core properties:
  - Continuous representation of objects and actions of interest
  - Physical actions and button pressing instead of issuing commands with complex syntax
  - Rapid reversible actions with immediate feedback on object of interest




E
2
1
M

Benefits of direct manipulation

- Novices can learn the basic functionality quickly
- Experienced users can carry out a wide range of tasks efficiently and rapidly
- Intermittent users can retain operational concepts over time
- Error messages rarely needed
- Users can immediately see if their actions are furthering their goals, and if not, do something else
- Users experience less anxiety
- Users gain confidence and mastery and feel in control

Disadvantages

- Some people take the metaphor of direct manipulation too literally
- Not all tasks can be described by objects, and not all actions can be done directly
- Some tasks are better achieved through delegating, for example, spell checking
- Can become screen space ‘gobblers’
- Moving a cursor using a mouse or touchpad can be slower than pressing function keys to do the same actions

4. Exploring

- Involves moving through virtual or physical environments
  - People can explore aspects of a 3D environment
  - Physical environments can also be embedded with sensors that when detect the presence of someone will trigger digital or physical events to happen
- Many examples of virtual environments, including cities, parks, buildings, rooms, and datasets
  - enable people to fly over them and zoom in and out of different parts

Seeing things larger than life and in situ in AR

A virtual lion appearing in someone’s living room created with Google 3D object

Exploring data in VR

Someone using Immersive data analytics allowing them to explore Tableau data in 3D

Responding*

- System takes the initiative to alert user to something that it "thinks" is of interest
- System does this by:
  - detecting the location and-or presence of someone in a vicinity and notifying them on their phone or smartwatch
  - what it has learned from their repeated behaviors
- Examples:
  - alerts the user of a nearby coffee bar where some friends are meeting
  - A wearer's fitness tracker notifies them of a milestone reached
- Automatic system response without any requests made by the user

* This type suggested by Christopher Lueg et al. (2018)

Potential cons of system-initiated notifications

- Can get tiresome or frustrating if too many notifications or the system gets it wrong
- What does it do when it gets something wrong?
  - Does it apologize?
  - Does it allow the user to correct the advice or information?

Choosing an interaction type

- Direct manipulation is good for ‘doing’ types of tasks, for example, designing, drawing, flying, driving, or sizing windows
- Issuing instructions is good for repetitive tasks, for example, spell-checking and file management
- Having a conversation is good for finding out information, requesting music/TV or learning
- Hybrid conceptual models are good for supporting multiple ways of carrying out the same actions

Difference between interaction types and interface styles

Interaction type:

- A description of what a person is doing when interacting with a system, for example, instructing, talking, browsing, or responding

Interface style:

- The kind of interface used to support the interaction, for instance, command, menu-based, gesture, or voice

Many kinds of interface styles available

- Command
- Speech
- Data-entry
- Form fill-in
- Query
- Graphical
- Web
- Pen
- Augmented reality
- Gesture

Other sources

Conceptual knowledge that is used to inform design and guide research include:

- Paradigms
- Challenges
- Visions
- Theories
- Models
- Frameworks

Paradigms

- Inspiration for a conceptual model
- General approach adopted by a community for carrying out research
  - Shared assumptions, concepts, values, and practices
  - For example, desktop, ubiquitous computing, in the wild

Examples of paradigms in HCI

- Ubiquitous computing
- Pervasive computing
- Wearable computing
- Internet of Things (IoT)
- Human-Centered Artificial Intelligence

Challenges

- Society is facing new challenges as the world changes
- Most well known are sustainable development goals (SDGs)
  - e.g. achieving zero hunger, quality education, gender equality and sustainable cities and communities
- HCI researchers are now addressing many societal challenges such as climate change
  - e.g. challenging the values designed into technologies that influence society's practices of disposal and recycling

Visions

- A driving force that frames research and development
- Invites people to imagine what life will be like in 10, 15, or 20 years' time
  - For example, Apple's 1987 knowledge navigator
  - Smart cities, smart health
  - Human-centered AI
- Provide concrete scenarios of how society can use the next generation of imagined technologies
- Also raise ethical questions such as, privacy and trust

Questions raised by tech visions

- How to enable people to access and interact with information in their everyday lives
- How to design user experiences where there is no obvious user control
- How and in what form to provide contextually-relevant information to people
- How to ensure that information passed around interconnected devices and objects is secure and protects people's privacy

Theory

- Explanation of a phenomenon
  - For example, information processing that explains how the mind, or some aspect of it, is assumed to work
- Can help identify factors relevant to the design and evaluation of interactive products
  - Such as cognitive, social, and affective
- Can be used to predict what users will do with different interfaces

Theory serves various roles in HCI

- descriptive - providing concepts and models
- explanatory – explicating relationships and processes
- predictive – testing hypotheses about user performance
- prescriptive – providing guidance for design
- generative – creating something new
- informative – selecting knowledge to couch our understanding
- conceptual – developing high-level frameworks
- critical – critiquing interaction design

Models

A simplification of an HCI phenomenon

- Enables designers to predict and evaluate alternative designs
- Abstracted from a theory coming from a contributing discipline, for example:
  - Don Norman's (1996) model of the Seven Stages of Action
  - Marc Hassenzahl's (2010) model of the user experience

Frameworks

- Set of interrelated concepts and-or specific questions for ‘what to look for’
- Provide advice on how to design user experiences
  - Helping designers think about how to conceptualize learning, working, socializing, fun, and emotion
- Focus on how to design particular kinds of interfaces to evoke certain responses
- Come in various forms:
  - Such as steps, questions, concepts, challenges, principles, tactics, and dimensions

A classic HCI framework

Don Norman's (1988) framework of the relationship between the design of a conceptual model and a user's understanding of it

Consists of three interacting components:

- *The Designer's Model*
  - The model the designer has of how the system should work
- *System Image*
  - How the system actually works, which is portrayed to the user through the interface, manuals, help facilities, and so on
- *The User's Model*
  - How the user understands how the system works

Summary

- Developing a conceptual model involves:
  - Understanding the problem space
  - Being clear about the underlying assumptions and claims
  - Specifying how the proposed design will support people in their activities
- A conceptual model is a high-level description of a product in terms of:
  - What users can do with it and the concepts they need to understand how to interact with it
- Interaction types provide a way of thinking about how to support user's activities
- Paradigms, challenges, visions, theories, models, and frameworks provide ways of framing design and research